

NANYANG TECHNOLOGICAL UNIVERSITY

CCDS25-0111

**END-TO-END AI SYSTEM FOR SINGAPORE REAL ESTATE: FROM
AUTOMATED DATA MINING TO AGENTIC RECOMMENDATION**

Submitted in Partial Fulfilment of the Requirements
for the Degree of Bachelor of Computing in Computer Science
of the Nanyang Technological University

by

Wang Yangming

Supervised by Prof. Zhang Jie

College of Computing and Data Science

2026

Abstract

This interim report details the development of an end-to-end AI-driven real estate analysis system tailored for the Singapore housing market. Addressing the industry-wide challenges of data fragmentation and the complex interplay between the public (HDB) and private residential sectors, this project aims to establish a unified market intelligence platform.

Significant progress has been made in the initial phase, specifically in data acquisition and infrastructure design. We have successfully designed and deployed robust web scraping pipelines for major property portals, including PropertyGuru, 99.co, EdgeProp, and SRX. These automated agents aggregate granular listing data—such as asking prices, floor areas, and tenure details—into a centralized relational database. A unified schema has been implemented to harmonize disparate data structures, incorporating preliminary algorithms for deduplication and cross-source linkage.

The report further delineates the strategic roadmap for the upcoming development cycles. Future work will prioritize the construction of a user-centric mobile application for property discovery and an administrative dashboard for data governance. Additionally, the project will transition towards advanced analytics, integrating Large Language Models (LLMs) to enable semantic search capabilities and predictive modeling for accurate property valuation. The deliverables achieved thus far provide a solid foundation for transforming raw market data into actionable, multi-dimensional real estate insights.

Acknowledgments

I would like to express my deepest gratitude to my supervisor, Prof. Zhang Jie, for his invaluable guidance, patience, and continuous support throughout the duration of this Final Year Project. His insightful feedback and profound knowledge in the field were instrumental in shaping the direction of this research and overcoming various technical challenges. I am truly grateful for the opportunity to work under his supervision.

I also wish to extend my sincere thanks to my project teammates, Darrell and Aaron. It has been a great pleasure working alongside such dedicated and talented individuals. Our collaboration, shared discussions, and mutual support were essential to the successful completion of this project. I appreciate the hard work we put in together and the camaraderie we shared.

Finally, I would like to thank the College of Computing and Data Science (CCDS) at Nanyang Technological University for providing the resources and environment necessary for this study.

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Chapter 1

Introduction

1.1 Background and Motivation

Singapore's housing sector is characterized by a unique **dual-tier structure**, comprising a highly regulated public housing market managed by the Housing and Development Board (HDB) and a free-market private residential sector. The HDB sector houses approximately 80% of the resident population, serving as a critical pillar of national social stability, while the private sector caters to higher-income households and investors [1]. Despite their regulatory separation, these two markets exhibit complex, non-linear interdependencies. The "asset progression" aspiration—whereby households upgrade from public to private housing—creates a dynamic flow of capital and demand that links the two tiers [2].

In the current digital era, the proliferation of online property portals such as PropertyGuru, 99.co, and SRX has led to an explosion of real estate data. This includes not only transactional records but also unstructured text descriptions, listing metadata, and visual assets. However, this data remains fragmented across disparate platforms, creating significant **information asymmetry** for stakeholders. Homebuyers, investors, and policymakers alike face challenges in obtaining a holistic view of the market, often relying on disjointed manual searches or simplistic linear analyses that fail to capture the nuances of the dual-market system.

1.2 Problem Statement

The core problem addressed by this research is the **lack of a unified, comprehensive, and intelligent analytics framework** for the Singapore real estate market. Specifically, three key challenges exist:

1. **Data Fragmentation:** Critical market intelligence is siloed across competing platforms, making cross-sectional analysis difficult.
2. **Methodological Limitations:** Traditional econometric methods (e.g., Hedonic Pricing Models) typically rely on structured variables and linear assumptions, often neglecting the rich insights embedded in unstructured data (listings descriptions, agent sentiment) and non-linear market dynamics.
3. **Regulatory Complexity:** Existing tools often fail to account for the granular impact of Singapore's frequent cooling measures and policy shifts on different market segments.

1.3 Project Objectives

This project aims to develop an end-to-end AI-driven system that bridges the gap between raw, fragmented data and actionable market intelligence. The specific objectives are:

1. **To Develop a Unified Data Infrastructure:** Design and implement automated web scraping pipelines to aggregate and harmonize data from Singapore's major property portals into a centralized repository.
2. **To Construct a Relational Graph of the Property Market:** Establish a robust database schema that links transient listings to persistent property master records (HDB and Condo), facilitating longitudinal tracking.
3. **To Implement Advanced Analytics:** Leverage machine learning techniques to extract semantic features from unstructured text and identify cross-market patterns.

4. **To Create Actionable User Interfaces:** Develop user-centric applications (Mobile App and Admin Dashboard) to democratize access to this unified intelligence.

1.4 Contributions

This study contributes to the field of Real Estate Informatics in the following ways:

- **Integration:** It proposes one of the first unified data architectures specifically designed for Singapore’s dual-tier market.
- **Methodology:** It demonstrates the feasibility of combining traditional structured data fields with unstructured text features for market analysis.
- **Application:** It provides a working prototype of a system that serves the needs of both data consumers (homebuyers) and data administrators.

1.5 Report Organization

The remainder of this report is organized as follows:

- **Chapter 2 (Literature Review)** surveys existing research on real estate valuation models, machine learning applications in property markets, and the specific characteristics of the Singapore housing context.
- **Chapter 3 (Methodology and System Design)** details the system architecture, data acquisition pipelines, and database design.
- **Chapter 4 (Implementation Progress)** documents the current status of the development, including the successful deployment of scrapers and the initial database population.
- **Chapter 5 (Future Work)** outlines the roadmap for the next phase of development, focusing on the frontend application and advanced AI modeling.

Chapter 2

Literature Review

2.1 Introduction

Real estate valuation and market analysis have long been subjects of academic and practical interest. This chapter surveys the evolution of methodologies in this domain, ranging from traditional econometric approaches to modern artificial intelligence applications. It also examines the specific literature regarding the Singapore housing market to contextualize the unique challenges addressed by this project.

2.2 Traditional Real Estate Valuation Models

The foundational framework for property valuation is the **Hedonic Pricing Model**, popularized by Rosen [3]. This theory posits that a heterogeneous good, such as a house, can be valued as the sum of its constituent attributes (e.g., location, size, number of bedrooms).

- **Strengths:** These models provide interpretability, allowing analysts to quantify the marginal value of specific features (e.g., the premium for an extra bathroom).
- **Limitations:** They often rely on strict assumptions of linearity and independence among variables. Furthermore, they struggle to incorporate high-dimensional or unstructured data, such as textual property descriptions or complex spatial

relationships.

2.3 Machine Learning in Real Estate

In recent years, the limitations of traditional models have led researchers to explore Machine Learning (ML) techniques.

- **Automated Valuation Models (AVMs):** Advanced regression techniques (e.g., Random Forests, Gradient Boosting) have been shown to outperform linear regression in price prediction accuracy by capturing non-linear relationships [4].
- **Natural Language Processing (NLP):** Recent studies, such as Baur et al. [5], utilize NLP to extract sentiment and semantic features from listing descriptions, demonstrating that "soft" factors significantly influence pricing.
- **Neural Networks:** Deep learning architectures allow for the integration of multi-modal data (text and images), providing a more holistic view of property value [6].

2.4 The Singapore Context: A Dual-Tier Market

Research on the Singapore market highlights its unique dual structure. Tu [1] analyzed the dynamics of the Singapore private housing market, finding significant relationships between private housing prices and the resale public housing sector. Sing et al. [2] further demonstrated price dynamics between public and private housing, where rising private prices can pull up HDB resale prices due to the substitution effect and inter-market mobility.

2.5 Research Gap

While existing literature covers individual components—valuation algorithms or market segregation—there is a paucity of research that integrates **unified data mining** across the dual-market spectrum with **modern NLP techniques**. Most local studies

rely on structured transaction data alone, ignoring the rich, unstructured listing data available on digital platforms. This project aims to bridge this gap by building a unified system that leverages both structured and unstructured data across the entire Singapore housing ecosystem.

Chapter 3

Methodology and System Design

3.1 System Architecture

The proposed system adopts a modular architecture consisting of five main stages: Data Acquisition, Data Processing, Data Storage, Analysis Engine, and Application Layer. The implementation focuses on building a robust data pipeline that aggregates property information from multiple sources into a unified repository.

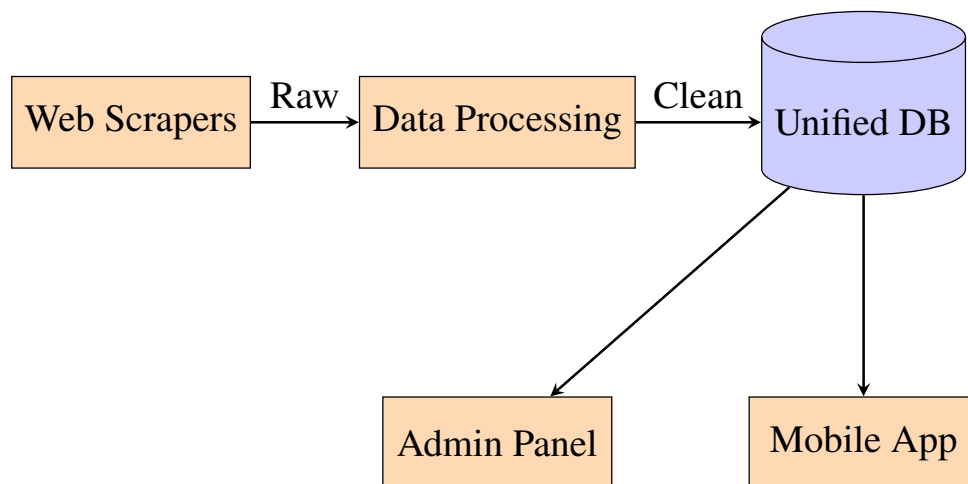


Figure 3.1: High-Level System Architecture Diagram

3.1.1 Data Acquisition

The system targets four major real estate portals in Singapore: PropertyGuru, 99.co, EdgeProp, and SRX. Specialized web scrapers are developed for each platform to extract granular data, including listing titles, prices, property attributes (size, bedrooms, tenure), and location details.

Table 3.1: Target Data Sources and Extraction Parameters

Platform	Key Data Attributes	Scraper Status
PropertyGuru	Pricing, Amenities, MRT Proximity	Operational
99.co	Tenure, Unit Size, Lease Date	Operational
EdgeProp	Historical Transactions, Trends	Operational
SRX	X-Value, Drone Views, Site Plans	In Development

The scrapers are designed to handle pagination and varying HTML structures across different sites.

3.1.2 Data Processing and Aggregation

Raw data fetched from different sources frequently contains inconsistencies and duplicates. The processing module performs the following tasks:

- **Normalization:** Standardizing field names and formats (e.g., converting "2 beds" and "2 Bedroom" to a standard integer format).
- **Deduplication:** Identifying and removing duplicate listings based on composite keys (e.g., address, unit size, and price) to ensure data integrity.
- **Linking:** Associating temporary rental/sale listings with persistent property master records (HDB or Condo units).

3.1.3 Data Storage

The data storage layer is built upon a PostgreSQL relational database, designed to maintain data integrity while supporting complex analytical queries.

Entity-Relationship Design

The database schema is centered around the separation of transient listing data from persistent property master data. Figure 3.2 illustrates the core entities and their relationships.

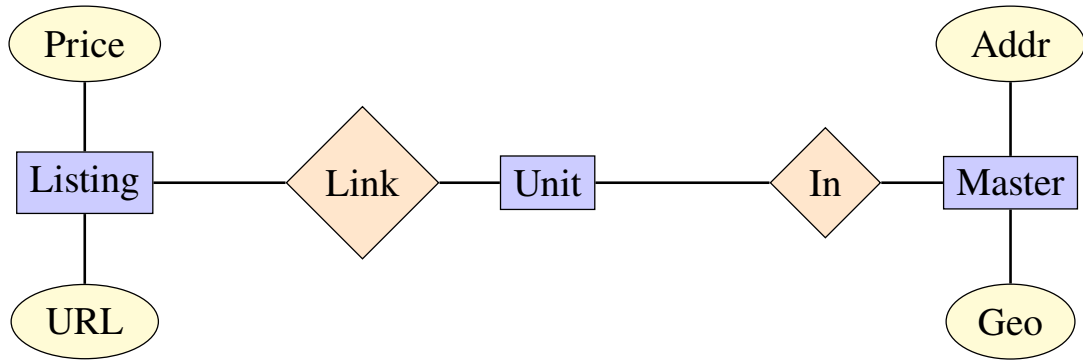


Figure 3.2: Entity-Relationship Diagram illustrating the linkage between scraped Listings and persistent Master records.

The `aggregated_listings` table serves as the primary repository for raw data, receiving inputs from all scrapers.

3.2 Database Design

A relational database schema is designed to store both static property information and dynamic listing data. The core tables are structured as follows:

3.2.1 Property Master Tables

To maintain a definitive registry of properties, we define master tables for Housing Development Board (HDB) flats and private condominiums:

1. **hdb_basic:** Stores block-level information for HDB estates (e.g., address, completion year, lease start date).
2. **hdb_unit:** Stores unit-level details, linking specific units to their respective blocks.

3. **condo_basic:** Stores project-level information for condominiums (e.g., project name, developer, facilities).
4. **condo_unit:** Stores specific unit details within each condominium project.

3.2.2 Listing Tables and Association

The `unified_listings` table aggregates active rental and sale listings from the web scrapers. A key design consideration is the association between these transient listings and the master tables.

- **Association Logic:** Since online listings provided by agents often lack specific unit numbers for privacy reasons, a direct foreign key constraint is not always possible during the scraping phase.
- **Agent-Assisted Mapping:** The system supports a "soft link" where the association between a listing and a specific physical unit (e.g., in `hdb_unit`) is finalized when an agent arranges a viewing or verifies the property documentation. This design allows for flexible data ingestion while enabling precise tracking as the transaction progresses.

Chapter 4

Implementation Progress

4.1 Data Mining and Scraper Development

Significant progress has been made in the Data Acquisition phase. We have successfully developed and deployed Python-based web scrapers for the target platforms.

4.1.1 Multi-Platform Scraping

Scripts have been implemented for PropertyGuru, 99.co, EdgeProp, and SRX. These scripts utilize libraries such as Selenium and BeautifulSoup to navigate search results, handle dynamic content loading, and extract structured data.

- **Coverage:** The scrapers are capturing listing descriptions, asking prices, floor areas, lease details, and geographical coordinates.
- **Robustness:** Error handling mechanisms have been integrated to manage connection timeouts and anti-scraping measures (e.g., CAPTCHA challenges, though manual intervention is sometimes required).

4.2 Database Implementation

The initial database infrastructure has been set up using SQLite for the development phase, with plans to migrate to PostgreSQL for production.

4.2.1 Unified Listing Repository

We have implemented point 1 of the requirements: merging and storing information in a unified structure. The raw data from different scrapers is harmonized into a single schema, facilitating cross-platform comparative analysis.

• 4.2.2 De-duplication and Record Linkage

Algorithmic Approach

A critical challenge in aggregating data from multiple portals is the presence of duplicate listings—agents often cross-post the same unit on PropertyGuru, 99.co, and SRX simultaneously. To address this, we developed a multi-stage de-duplication algorithm:

1. **Address Normalization:** Raw address strings are first passed through a cleaning pipeline. This involves:
 - Case standardization (lowercasing).
 - Abbreviation expansion (e.g., "Ave" → "Avenue", "Blk" → "Block").
 - Removal of non-alphanumeric noise characters.
2. **Blocking Strategy:** To avoid an exponential $O(N^2)$ comparison, records are "blocked" by their postal code prefixes. Comparisons are only performed within each block, significantly reducing computational overhead.
3. **Fuzzy Matching Logic:** Within each block, a pairwise similarity score is calculated using the **Levenshtein Distance** metric on the listing titles and normalized addresses. We also employ a "Sliding Window" temporal filter: two listings are only considered potential duplicates if they were posted within a 7-day window.
4. **Consensus Resolution:** When duplicates are identified (Similarity Score > 0.85), they are merged into a single "Golden Record". The system prioritizes data from PropertyGuru for pricing fields (due to higher up-

date frequency) and 99.co for tenure information (due to better structured metadata).

- **Schema:** The `aggregated_listings` table is live and populated with sample datasets from all four sources.

4.3 Current Limitations

While the data collection is functional, the exact mapping of listings to specific HDB/Condo units (Points 2 & 3) is currently limited by the lack of visible unit numbers in public listings. The infrastructure for the master tables (`hdb_unit`, `condo_unit`) exists, but population is ongoing.

Chapter 5

Future Work plan

The next phase of the project will focus on Application Development and Advanced Analysis, addressing Points 4 and 5 of the project scope.

5.1 Frontend Development

5.1.1 Mobile Application

A mobile-responsive frontend will be developed to allow users to interact with the aggregated database.

- **Search Functionality:** Users will be able to input specific query criteria (e.g., district, price range, floor area) and retrieve a filtered list of listings.
- **Aggregated View:** The app will display the total number of matching listings across all platforms, providing a comprehensive market overview that single-platform apps cannot offer.

5.2 Backend and Administration

5.2.1 Admin Dashboard

A backend maintenance interface will be built to support the Create, Read, Update, and Delete (CRUD) operations required for system management.

- **Master Data Management:** Administrators will be able to maintain the `hdb_basic` and `condo_basic` master tables, ensuring new projects and blocks are added.
- **Data Cleaning:** The interface will allow manual correction of scraper errors and the merging of duplicate records that the automated algorithm misses.

5.3 Advanced AI Features

Building upon the solid data foundation, we plan to implement the AI-driven features outlined in the introduction.

- **Semantic Search:** Leveraging LLMs to allow users to search using natural language (e.g., "spacious high-floor unit near MRT") rather than rigid filters.
- **Price Recommendation:** Training a model on historical transaction data and current listing data to provide estimated fair values for specific units.

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